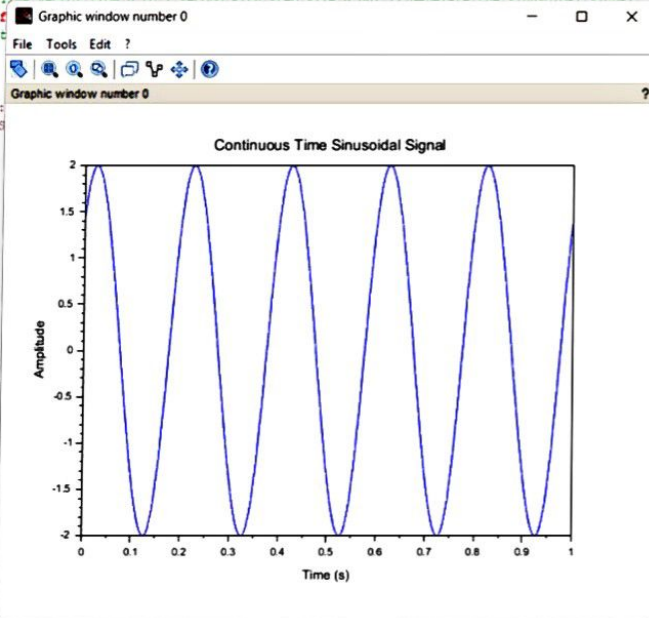




```
1 // Test Case: Sinusoidal Signal Generation // Define parameters
2 f = 5; // Frequency in Hz
3 A = 2;
4 phi = pi/4; // Amplitude - Phase shift in radians
5 t = 0:0.001:1; // Time vector from 0 to 1 second with step size 0.001 // Generate the sinusoidal signal
6 x_t = A * sin(2 * pi * f * t + phi);
7 clf; // Clear the current figure
8 plot(t, x_t);
9 xlabel('Time (s)');
10 ylabel('Amplitude');
11 title('Sinusoidal Signal');
12 xtitle('Continuous-Time Sinusoidal Signal');
13
```





```
1 // Test Case: Discrete-Time Unit Impulse Signal Generation
2 // Define parameters
3 N = 21; // Length of the signal
4 n0 = 11; // Discrete-Time Index of the impulse
5 n = 1:N;
6 // Generate the unit impulse signal
7 x_n = zeros(1, N); // Initialize signal with zeros
8 x_n(n0) = 1; // Set the impulse at the desired location
9 // Plot the signal
10 clf; // Clear the current figure
11 plot2d(n, x_n);
12 xlabel('n-(Discrete Time Index)');
13 ylabel('Amplitude');
14 title('Discrete-Time Unit Impulse Signal');
15 xtitle('Discrete-Time Unit Impulse Signal','n-(Discrete Time Index)','Amplitude');
16
```

